

Week	Focus	What he should learn	Deliverable
1	Serving-layer fundamentals	Difference between warehouse, semantic layer, API layer, and BI dashboard. Why APIs should expose domain operations, not raw tables.	One-page API design proposal: consumers, use cases, endpoints, request/response examples.
2	API contract design	REST basics, OpenAPI, Pydantic models, request validation, error format, versioning, pagination.	FastAPI skeleton with <code>/health</code> , <code>/metadata</code> , OpenAPI docs, typed request/response models.
3	Warehouse query integration	BigQuery/Python client or warehouse adapter pattern, parameterized SQL, SQL template organization, query timeout handling.	<code>/listings/search</code> endpoint backed by warehouse query templates. Use only allowlisted filters.
4	Query safety and cost control	No arbitrary SQL. Use parameterized queries, required date filters, column projection, row limits, dry runs, and max bytes billed. BigQuery parameters help protect against SQL injection, but they cannot replace table names, column names, or other identifiers, so identifiers must be allowlisted separately. (Google Cloud Documentation)	Add <code>QueryValidator</code> : allowed dimensions, allowed metrics, max limit, required snapshot/date filters.
5	Performance	Avoid <code>SELECT *</code> , select only needed columns, partition/date filtering, materialized gold tables, caching, pagination, async job pattern for expensive queries. BigQuery docs explicitly recommend controlling projection and querying only needed columns; <code>LIMIT</code> is not a cost-control mechanism for non-clustered tables. (Google Cloud Documentation) (Google Cloud Documentation)	Add <code>/market/metrics</code> endpoint using pre-aggregated gold table or materialized result table.

6	Auth and access control	API keys vs JWT, service accounts, least privilege, read-only warehouse access, per-client rate limits. API Gateway can validate JWTs before forwarding requests to the backend. (Google Cloud Documentation)	Protect all non-health endpoints. Add client identity to logs.
7	Deployment and reliability	Docker, Cloud Run deployment, environment config, secrets, request timeout, concurrency, autoscaling behavior. Cloud Run scales revisions based on incoming requests and lets you tune maximum concurrent requests per instance, which affects latency and cost. (Google Cloud Documentation)	Deploy to Cloud Run with staging/prod configs and basic CI/CD.
8	Observability and handoff	Structured logs, request IDs, latency metrics, warehouse query stats, errors, tracing, API docs for AI team. OpenTelemetry has FastAPI instrumentation for HTTP request tracing. (opentelemetry-python-contrib.readthedocs.io)	Final demo: AI team can call the API using docs, sample Python client, and test tokens.